STPM 2025 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ (unless the question gives another value) and the values of constants printed in the question paper.

Section A [15 marks]

1. Which expression has a unit of kg m^2 s^{-3}?

A Mass × acceleration
B Current × voltage
C Force × distance
D Current × resistance

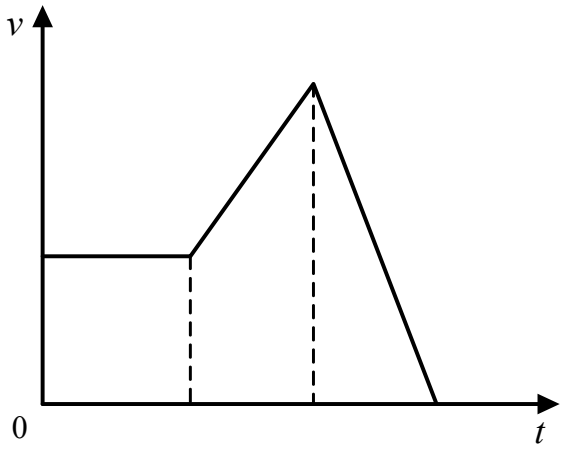
Answer: B

kg m^2 s^{-3} = (kg m s^{-2})(m)(s^{-1}) = J s^{-1} = W, the unit of power.

Current × voltage = IV = power (W)

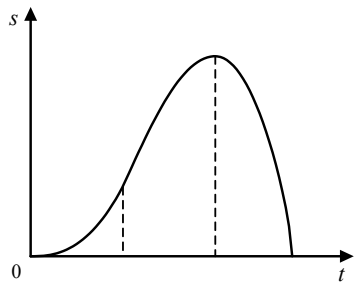
Explanation: Check the unit of each option: mass × acceleration is a force (kg m s^{-2}), force × distance is work (kg m^2 s^{-2}) and current × resistance is a voltage. Only IV is a power, whose base unit is kg m^2 s^{-3}.

2. The variation of velocity, v, with time, t, of a car is shown in the graph below.

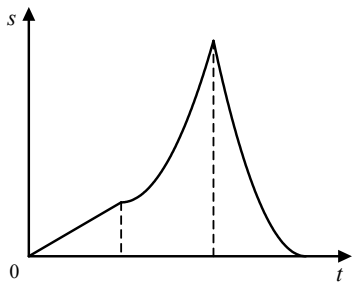


Which graph represents the variation of the displacement, s, with time, t, of the car?

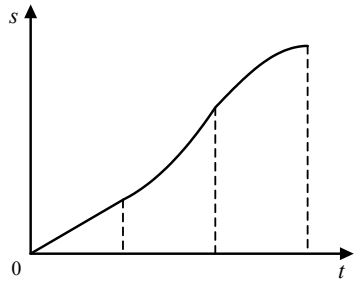
A



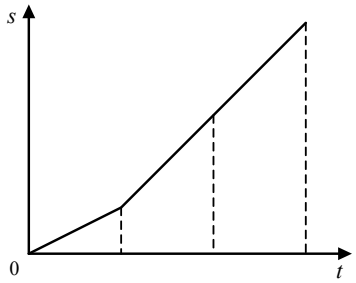
B



C



D



Answer: C

Displacement = area under the v–t graph, and the gradient of the s–t graph = velocity.

• Constant velocity: s increases linearly (straight line).

• Velocity increasing: the s–t graph becomes steeper (curves upward).

• Velocity decreasing to zero: the s–t graph becomes less steep and is horizontal when $v = 0$.

Explanation: The velocity is never negative, so the displacement never decreases: graphs A and B, which come back down, are wrong. Graph D uses only straight lines, which would mean the velocity jumps between constant values. Only C has a straight section, then a section that curves upward, then one that levels off.

3. Which is **not** true about the definition of Newton's laws of motion?

A The rate of change of the momentum of an object is directly proportional to the external force that acts on the object and in the same direction with the acceleration.

B Every object remains in the state of rest or uniform motion in a straight line unless when there is an external force that acts on the object.

C The total momentum of a system of interacting objects remains constant unless when there is an external force that acts on the object.

D If object P exerts force on object Q, then object Q exerts an equal and oppositely-directed force on object P.

Answer: C

Statement C is the principle of conservation of linear momentum. It follows from Newton's laws but it is not the definition of any of Newton's three laws.

A is the second law, B is the first law and D is the third law.

Explanation: Learn the three laws in their standard wording: (1) a body stays at rest or in uniform motion in a straight line unless an external force acts on it; (2) the rate of change of momentum is directly proportional to the resultant force and takes place in the direction of the force; (3) action and reaction are equal in magnitude, opposite in direction and act on different bodies.

4. A ball of mass 0.140 kg is released from a height of 5.20 m. If the ball rebounds with the same speed during impact, what is the impulse of the ball?

A 2.67 N s B 2.75 N s C 2.83 N s D 10.1 N s

Answer: C

$$v = \sqrt{(2gh)} = \sqrt{(2(9.81)(5.20))} = 10.1 \text{ m s}^{-1}$$

$$\text{Impulse} = \Delta p = mv - (-mv) = 2mv = 2(0.140)(10.1) = 2.83 \text{ N s}$$

Explanation: Impulse equals the change in momentum. Momentum is a vector: the ball arrives downwards and leaves upwards with the same speed, so the change is $2mv$, not zero and not mv. Option D (10.1) is only the speed of the ball.

5. An object of mass 0.80 kg which is tied to one end of a string of length 2.0 m swings as a pendulum. At the lowest point of swing, the object has maximum energy of 10 J. What is the speed of the object at the instant the angle of the string is 50° with the vertical?

A 3.3 m s^{-1} B 4.4 m s^{-1} C 5.0 m s^{-1} D 5.6 m s^{-1}

Answer: A

$$h = l(1 - \cos 50°) = 2.0(1 - \cos 50°) = 0.714 \text{ m}$$

$$E_p = mgh = (0.80)(9.81)(0.714) = 5.61 \text{ J}$$

$$\frac{1}{2}mv^2 = 10 - 5.61 = 4.39 \text{ J} \Longrightarrow v = \sqrt{\left(\frac{2(4.39)}{0.80}\right)} = 3.3 \text{ m s}^{-1}$$

Explanation: The total mechanical energy of the pendulum (10 J) is constant. At the lowest point it is all kinetic energy. When the string is at 50° to the vertical the bob has risen by $l - l \cos 50°$, so part of the 10 J is now potential energy and the rest is kinetic energy.

6. A car of mass 1000 kg travels on a humped bridge of radius 20 m with constant speed of 8.0 m s^{-1}. What is the force exerted by the car on the road surface when the car is at the highest position of the bridge?

[Assume that the gravitational acceleration is 10 m s^{-2}.]

A 3200 N B 6800 N C 10 000 N D 13 200 N

Answer: B

At the top of the bridge the resultant downward force provides the centripetal force:

$$mg - R = \frac{mv^2}{r}$$

$$R = (1000)(10) - \frac{(1000)(8.0)^2}{20} = 10\,000 - 3200 = 6800 \text{ N}$$

By Newton's third law the car pushes on the road with 6800 N.

Explanation: At the top of a hump the centre of the circle is below the car, so the resultant force must point downwards: the weight is larger than the normal reaction. The car therefore presses on the road with less than its weight. Option A (3200 N) is only the centripetal force; option C is the weight.

7. The mass of the Earth is 6.00×10^{24} kg and the mass of the Moon is 7.40×10^{22} kg. The distance between the Earth and the Moon is 3.80×10^{8} m. What is the force of attraction between the Earth and the Moon?

A 1.17×10^{39} N B 3.07×10^{30} N C 7.79×10^{28} N D 2.05×10^{20} N

Answer: D

$$F = \frac{GMm}{r^2} = \frac{(6.67 \times 10^{-11})(6.00 \times 10^{24})(7.40 \times 10^{22})}{(3.80 \times 10^{8})^2} = 2.05 \times 10^{20} \text{ N}$$

Explanation: Newton's law of gravitation uses the square of the separation. Option C comes from forgetting to square r; take care with the powers of ten when you use a calculator.

8. A satellite is orbiting around the Earth in a circular orbit. Owing to less air resistance, the satellite drops to a lower orbit. If the total energy of the satellite decreases by ΔE, what is the changes in the kinetic and potential energies?

	Kinetic energy	*Potential energy*
A	Increases by ΔE	Remains constant
B	Increases by ΔE	Decreases by $2\Delta E$
C	Decreases by $\frac{1}{2}\Delta E$	Decreases by $\frac{1}{2}\Delta E$
D	Remains constant	Decreases by ΔE

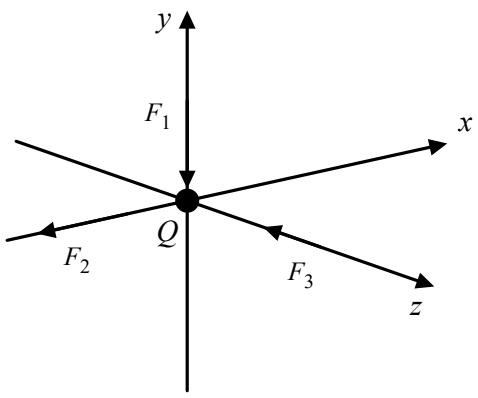
Answer: B

$$E_k = \frac{GMm}{2r},\ U = -\frac{GMm}{r},\ E = -\frac{GMm}{2r}$$

A lower orbit (smaller r) makes E more negative by ΔE. Since $E_k = -E$ and $U = 2E$: E_k increases by ΔE and U decreases by $2\Delta E$.

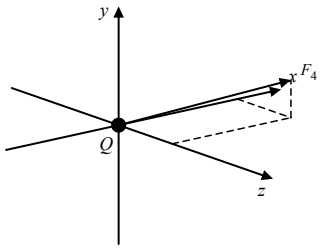
Explanation: This is a surprising result: air resistance makes the satellite speed *up*. As it spirals inwards it loses potential energy twice as fast as it loses total energy, and half of the potential energy lost becomes extra kinetic energy (the other half is lost to air resistance).

9. Three forces, F_1, F_2 and F_3, that act on a small sphere, Q, is shown in the Cartesian diagram below.

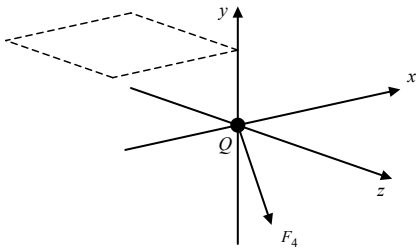


If $F_1 < F_2 < F_3$, which Cartesian diagram shows the correct direction of another force, F_4, that acts on Q and Q becomes stationary?

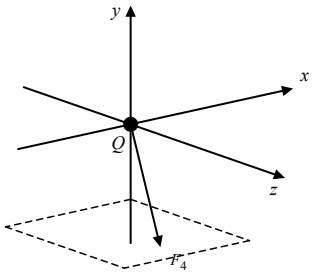
A



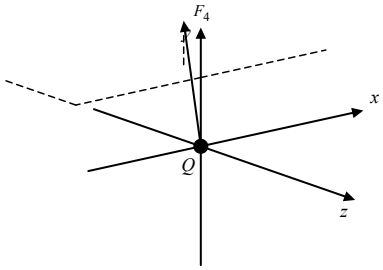
B



C



D



Answer: A

F_1 acts along $-y$, F_2 along $-x$ and F_3 along $-z$. For equilibrium, $F_4 = -(F_1 + F_2 + F_3)$, so F_4 has positive x, y and z components, with the z component largest and the y component smallest.

Explanation: For a particle to be stationary the vector sum of all the forces must be zero, so the fourth force must cancel the resultant of the other three. Its components are equal in size and opposite in direction to the sum of the given components. Only diagram A has a force pointing above the xz-plane, between the positive x and z axes.

10. Which statement is true about copper and rubber?

A Copper and rubber have similar stress-strain curve.
B The molecular chains in copper are coiled and tangled.
C Rubber exhibits loading-unloading cycle in stress-strain graph.
D Copper extends more than rubber when the same amount of force is applied.

Answer: C

Rubber shows a loading–unloading loop (hysteresis): the unloading curve lies below the loading curve, and the area between them is the energy converted to heat in one cycle.

Explanation: Copper is a ductile metal: its atoms are arranged in a crystal lattice, it obeys Hooke's law for small strains and then deforms plastically. Rubber is a polymer whose long molecular chains are coiled and tangled (so B describes rubber, not copper); the chains straighten out when stretched, so rubber extends much more than copper for the same force.

11. An energy, E, of an ideal gas at temperature T is given by $\frac{5}{2}RT$. E represents

A the total kinetic energy of one mole of gas
B the total kinetic energy of one molecule of gas
C the translational kinetic energy of one mole of gas
D the translational kinetic energy of one molecule of gas

Answer: A

$$E = \frac{f}{2}RT \text{ per mole; } f = 5 \text{ for a diatomic gas (3 translational + 2 rotational)}$$

So $\frac{5}{2}RT$ is the total kinetic energy of one mole of a diatomic gas.

Explanation: By the principle of equipartition each degree of freedom has $\frac{1}{2}kT$ per molecule or $\frac{1}{2}RT$ per mole. The translational kinetic energy is always $\frac{3}{2}RT$ per mole ($\frac{3}{2}kT$ per molecule). The factor R (not k) shows the energy is per mole.

12. Which is true about the relationship between the Boltzmann constant, k, and the molar gas constant, R?

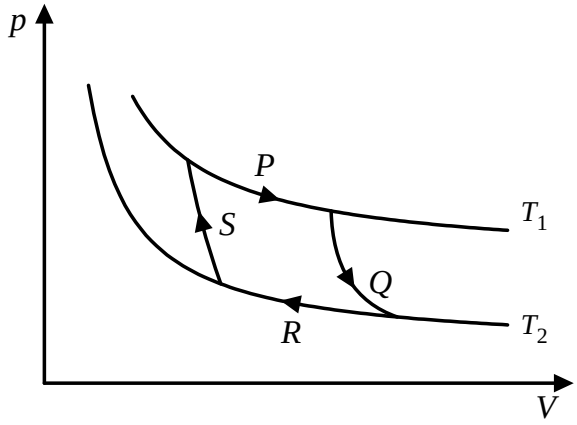
A $\frac{R}{k} = 6.02 \times 10^{23}$
B $Rk = 6.02 \times 10^{23}$
C $\frac{k}{R} = 6.02 \times 10^{23}$
D $R + k = 6.02 \times 10^{23}$

Answer: A

$$R = N_A k \Longrightarrow \frac{R}{k} = N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

Explanation: The Boltzmann constant is the gas constant per molecule: $k = \frac{R}{N_A} = \frac{8.31}{6.02 \times 10^{23}} = 1.38 \times 10^{-23} \text{ J K}^{-1}$.

13. A cyclic thermodynamic process, *PQRS*, is shown in a *p*-*V* graph below.



Which statement is **not** true about the graph?

A The gas expands adiabatically along *Q*.
B The gas compresses isothermally along *P*.
C The heat transferred in the system is zero along *Q* and *S*.
D The temperature of T_1 is higher than the temperature of T_2.

Answer: B

P lies on the isotherm T_1 and the volume increases: it is an isothermal **expansion**, not a compression, so B is not true.

Q and *S* are the steeper curves (adiabatic, $Q = 0$); *R* is an isothermal compression at T_2; the upper isotherm T_1 is the hotter one.

Explanation: On a *p*–*V* graph an adiabatic curve is steeper than an isotherm. The isotherm further from the origin is at the higher temperature ($pV = nRT$). Read the arrow on each process to decide whether the gas expands (volume increases) or is compressed.

14. A metal block of density 21.6 g cm^{-3} and volume 5.60 cm^3 has heat capacity of 38.4 J K^{-1}. What is the specific heat capacity of the metal block?

A 26.0 J kg^{-1} K^{-1} B 99.5 J kg^{-1} K^{-1} C 148 J kg^{-1} K^{-1} D 317 J kg^{-1} K^{-1}

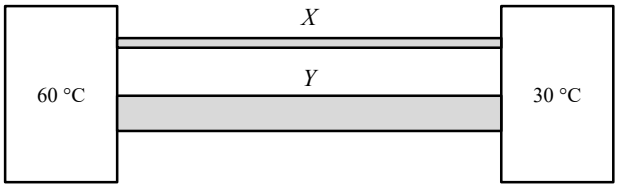
Answer: D

$$m = \rho V = (21.6)(5.60) = 121 \text{ g} = 0.121 \text{ kg}$$

$$c = \frac{C}{m} = \frac{38.4}{0.121} = 317 \text{ J kg}^{-1} \text{ K}^{-1}$$

Explanation: Heat capacity *C* belongs to the whole block; specific heat capacity is per kilogram. Find the mass from the density first, and convert grams to kilograms before dividing.

15. Two perfectly insulated metal rods, X and Y, of the same length, 50 cm, are used to connect two reservoirs of temperatures 60 °C and 30 °C as shown in the diagram below.



The cross-sectional area of X and Y are 1.0×10^{-4} m^2 and 4.0×10^{-4} m^2 respectively. What is the rate of heat flow between both reservoirs at steady state?
[Thermal conductivities of X and Y are 25 W m^{-1} K^{-1} respectively.]

A 0.15 W B 0.38 W C 0.60 W D 0.75 W

Answer: D

Both rods connect the same two reservoirs, so they conduct side by side (in parallel):

$$P_X = \frac{kA\Delta\theta}{l} = \frac{(25)(1.0 \times 10^{-4})(30)}{0.50} = 0.15 \text{ W}$$

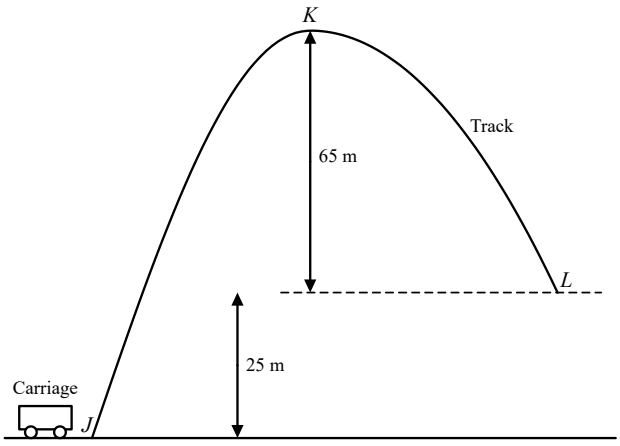
$$P_Y = \frac{(25)(4.0 \times 10^{-4})(30)}{0.50} = 0.60 \text{ W}$$

$$P = 0.15 + 0.60 = 0.75 \text{ W}$$

Explanation: Each rod has the full 30 K temperature difference across its 0.50 m length, and heat flows through both at the same time, so the total rate is the sum. Options A and C are the rates through one rod only.

Section B [15 marks]

16. A carriage of mass 500 kg is being pulled along a track from point J to point K by an electric motor as shown in the diagram below.



At point K, the carriage is released and slides down the track towards point L. The efficiency of the motor is 60%.

(a) Calculate the input power of the motor if the carriage takes 30 s to move from point J to point K. [3]

Answer

Height of K above $J = 25 + 65 = 90$ m; $E_p = mgh = (500)(9.81)(90) = 4.41 \times 10^5$ J — 1

Output power $= \frac{4.41 \times 10^5}{30} = 1.47 \times 10^4$ W; efficiency $= \frac{\text{output power}}{\text{input power}}$ — 1

Input power $= \frac{1.47 \times 10^4}{0.60} = 2.45 \times 10^4$ W — 1

3

Explanation: The useful work done by the motor raises the carriage from J to K; read both heights from the diagram (K is 65 m above L, and L is 25 m above J). Only 60% of the input power is useful, so the input power is larger than the output power.

(b) Calculate the speed of the carriage at point L. [3]

Answer

Loss of E_p from K to L = gain of E_k: $mgh = \frac{1}{2}mv^2$ — 1

$v = \sqrt{(2(9.81)(65))}$ — 1

$v = 35.7$ m s^{-1} — 1

[Assume the carriage starts from rest at K and the track is frictionless]

3

Explanation: The carriage falls through only 65 m between K and L. The mass cancels, so the speed does not depend on the mass of the carriage.

(c) State a factor which affects the efficiency of the motor. [1]

Answer

Any one: friction in the moving parts / heat produced in the coil (resistance) of the motor / energy lost as sound. — 1

1

Total for question **7**

Explanation: The efficiency is less than 100% because part of the input energy is converted into heat and sound instead of useful work. Anything that increases these losses lowers the efficiency.

17. The temperature of helium gas in a container is 298 K. Assume that the helium gas behaves like an ideal gas.

[Given that the mass of helium atom is 6.646×10^{-27} kg.]

(a) Calculate the average translational kinetic energy of the helium gas. [2]

Answer

$E_k = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23})(298)$ — 1

$E_k = 6.17 \times 10^{-21}$ J — 1

2

Explanation: The average translational kinetic energy of a gas molecule depends only on the temperature: $\frac{3}{2}kT$. Helium is monatomic, so this is also its total kinetic energy.

(b) Calculate the root mean square speed of the helium gas. [3]

Answer

$\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT \Longrightarrow c_{\text{rms}} = \sqrt{(\frac{3kT}{m})}$ — 1

$c_{\text{rms}} = \sqrt{(\frac{3(1.38 \times 10^{-23})(298)}{6.646 \times 10^{-27}})}$ — 1

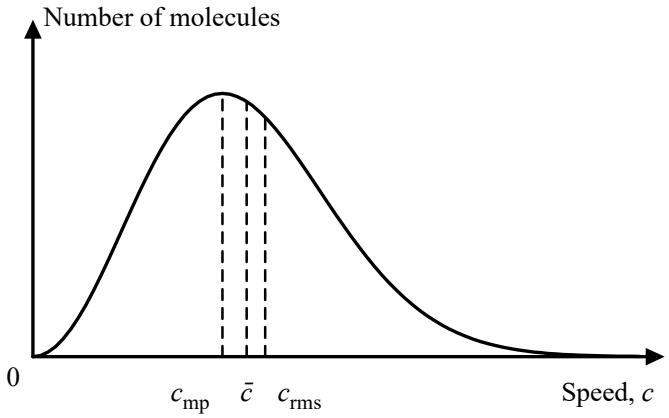
$c_{\text{rms}} = 1.36 \times 10^3$ m s^{-1} — 1

3

Explanation: The root mean square speed is the square root of the mean of the squares of the speeds. Use the mass of one atom (not the molar mass) with the Boltzmann constant k; or use the molar mass with R. Both give the same answer.

(c) Sketch the speed distribution graph of the helium gas. [3]

Answer



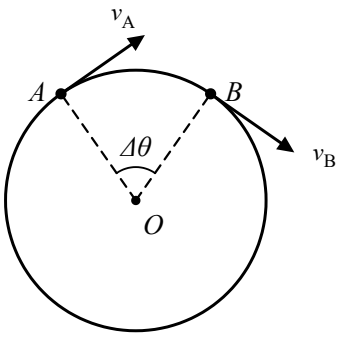
Axes labelled: number of molecules against speed	1
Curve starts from the origin, rises to a peak and falls with a long tail towards high speed (not symmetrical)	1
c_{mp} at the peak, with $\bar{c}$ and c_{rms} to its right in that order	1
	3
Total for question	**8**

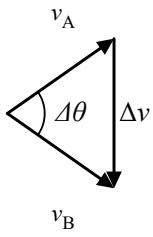
Explanation: This is the Maxwell–Boltzmann distribution. No molecule is at rest, most molecules have speeds near the most probable speed, and a few move very fast. Because of the long tail the three typical speeds are different: $c_{mp} < \bar{c} < c_{rms}$.

Section C [30 marks]

18. (a) Show that a particle that moves in uniform speed experiences an acceleration. [5]

Answer



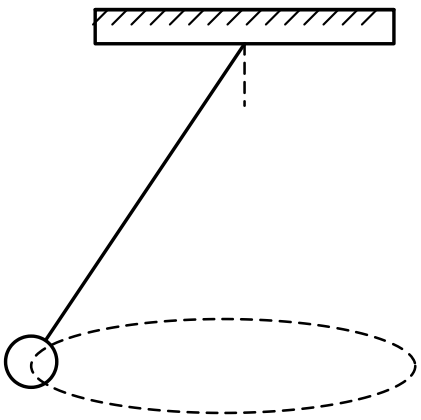


Δv points towards O

Velocity is a vector; its direction is along the tangent and keeps changing, although its magnitude (speed) is constant	1
From A to B the velocity turns through $\Delta\theta$; the change in velocity $\Delta v = v_B - v_A \neq 0$ (vector triangle)	1
For small $\Delta\theta$: $\Delta v = v\Delta\theta$	1
$a = \frac{\Delta v}{\Delta t} = v\frac{\Delta\theta}{\Delta t} = v\omega$	1
$a = \frac{v^2}{r}$ $(= r\omega^2)$, directed towards the centre	1
	5

Explanation: Acceleration is the rate of change of velocity, not of speed. Moving round a circle, the speed stays the same but the direction of motion keeps changing, so the velocity changes and there must be an acceleration. The vector triangle shows that the change in velocity points towards the centre of the circle.

(b) The bob of a conical pendulum moves in a horizontal circle as shown in the diagram below.



T is the tension in the light string and W is the weight of the bob. Derive an expression for the centripetal acceleration in terms of T, W and g. [5]

Answer

Let the string make an angle θ with the vertical. Vertical: $T\cos\theta = W$ — 1

Horizontal: $T\sin\theta = ma$ — 1

$m = \frac{W}{g}$ — 1

$\sin\theta = \sqrt{(1-\cos^2\theta)} = \frac{\sqrt{(T^2 - W^2)}}{T}$ — 1

$a = \frac{T\sin\theta}{m} = \frac{g\sqrt{(T^2 - W^2)}}{W}$ — 1

5

Explanation: The bob moves in a horizontal circle, so there is no vertical acceleration: the vertical component of the tension balances the weight. The horizontal component of the tension, pointing to the centre of the circle, is the centripetal force. Eliminate the angle and the mass to express a in terms of T, W and g.

(c) A load of mass 0.50 kg is tied to one end of a light string and swirled in a vertical circle of radius 1.5 m. The velocities of the load at the instant the load is at the highest and lowest points of the circle are 5.9 m s^{-1} and 9.7 m s^{-1} respectively. Determine the tension in the string when the load is at

(i) the highest point of the circle, [3]

Answer

At the highest point: $T + mg = \frac{mv^2}{r}$ — 1

$T = \frac{(0.50)(5.9)^2}{1.5} - (0.50)(9.81)$ — 1

$T = 6.70$ N — 1

3

Explanation: At the top of the circle both the tension and the weight point downwards, towards the centre, and together they provide the centripetal force. The tension is therefore smallest at the top.

(ii) the lowest point of the circle. [2]

Answer

At the lowest point: $T - mg = \frac{mv^2}{r} \Rightarrow T = \frac{(0.50)(9.7)^2}{1.5} + (0.50)(9.81)$ 1

$T = 36.3$ N 1

2

Total for question **15**

Explanation: At the bottom the tension points up (towards the centre) and the weight points down, so the tension must be larger than the weight by the centripetal force. The string is most likely to break here.

19. (a)

(i) State **two** differences between the centre of mass and the centre of gravity. [2]

Answer

The centre of mass is the point where the whole mass of a body can be considered to be concentrated; the centre of gravity is the point where the whole weight of the body appears to act. 1

The centre of mass does not depend on the gravitational field; the centre of gravity depends on the gravitational field (it exists only in a gravitational field). 1

2

Explanation: The two points are defined by different quantities: mass (a property of the body) and weight (a force that depends on g). A body in deep space still has a centre of mass but has no weight.

(ii) State the condition when the centre of mass is equal to the centre of gravity of a body. [1]

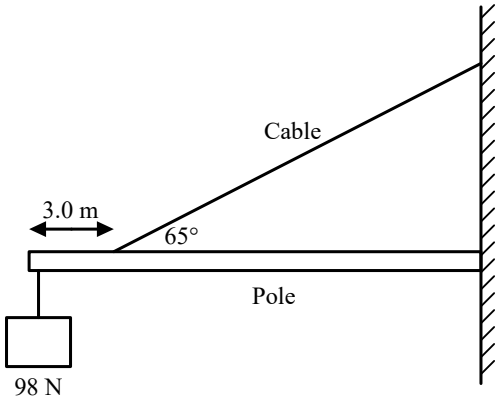
Answer

When the body is in a uniform gravitational field (g is the same at every point of the body). 1

1

Explanation: If every particle of the body has the same g, its weight is distributed exactly like its mass, so the two points coincide. This is true for everyday objects on the Earth.

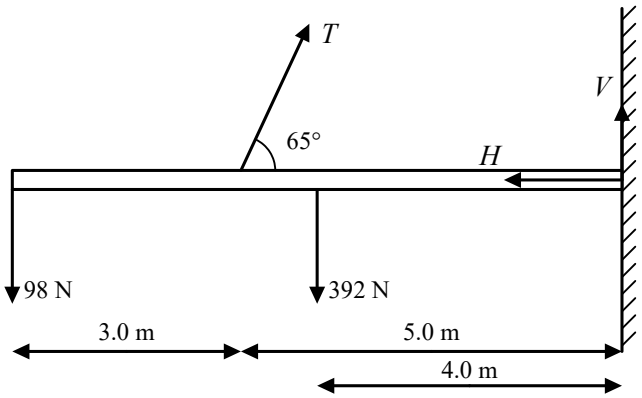
(b) A pole is hinged horizontally on a wall by a cable that makes an angle of 65° as shown in the diagram below.



A block of weight 98 N is suspended at the end of the pole 3.0 m away from the fastened cable on the pole. The weight and length of the pole are 392 N and 8.0 m respectively.

(i) Sketch a diagram that shows all the forces that act on the pole. [3]

Answer



Tension T along the cable at 65° to the pole, acting at 3.0 m from the free end	1
Weight of the pole, 392 N, at the centre and the load, 98 N, at the free end	1
Reaction of the wall at the hinge (or its components H and V)	1
	3

Explanation: Draw only the forces that act on the pole. The hinge can push or pull in any direction, so show the reaction at the wall as one force or as horizontal and vertical components.

(ii) Calculate the tension of the cable. [3]

Answer

Take moments about the hinge (the reaction has no moment):

$(T \sin 65°)(5.0) = (392)(4.0) + (98)(8.0)$ 1

$T \sin 65° = 470.4$ N 1

$T = 519$ N 1

3

Explanation: Choose the hinge as the pivot so that the unknown reaction drops out. Only the vertical component of the tension, $T \sin 65°$, has a moment about the hinge, and its distance from the wall is $8.0 - 3.0 = 5.0$ m. The weight of a uniform pole acts at its centre, 4.0 m from the wall.

(iii) Determine the magnitude and direction of the reaction that acts on the pole by the wall. [6]

Answer

Horizontal: $H = T \cos 65°$ 1

$H = 519 \cos 65° = 219$ N 1

Vertical: $V + T \sin 65° = 392 + 98$ 1

$V = 490 - 470.4 = 19.6$ N 1

$R = \sqrt{(219^2 + 19.6^2)} = 220$ N 1

Direction: $\tan^{-1} \frac{19.6}{219} = 5.1°$ above the horizontal, pointing away from the wall 1

6

Total for question **15**

Explanation: For equilibrium the resultant force is zero in every direction. The horizontal component of the tension pulls the pole towards the wall, so the wall pushes it away from the wall. Vertically, the tension supports most of the weight and the hinge provides the small remainder upwards.

20. (a) Explain the mechanism of heat conduction. [3]

Answer

When one end is heated, the atoms there vibrate with larger amplitude (more kinetic energy) 1

They pass energy to neighbouring atoms through the interatomic bonds/collisions, so energy flows from the hot region to the cold region without bulk movement of the material 1

In metals, free electrons gain kinetic energy, move quickly to the colder region and transfer energy by colliding with the lattice 1

3

Explanation: Conduction transfers energy through a material while the material itself stays in place. Metals conduct much better than non-metals because their free electrons carry energy quickly in addition to the lattice vibrations.

(b) An igloo is built in the shape of a hemisphere, with an outer radius of 2.80 m and 0.50 m thick walls of compacted snow. The compacted snow acts as insulator to trap the heat of a body inside the igloo. The thermal conductivity of a compacted snow is 0.15 W m^{-1} K^{-1}.

(i) If the body inside the igloo has temperature of 37 °C and it is a black body of surface area 1.7 m^2, calculate the power radiated by the body. [2]

Answer

$T = 37 + 273 = 310$ K; $P = \sigma AT^4 = (5.67 \times 10^{-8})(1.7)(310)^4$ 1

$P = 890$ W 1

[Accept 892 W using 310.15 K]

2

Explanation: For a black body the Stefan–Boltzmann law gives the power emitted, $P = \sigma AT^4$, with the temperature in kelvin.

(ii) If 40% of the heat radiated per second by the body is transferred through the wall of the igloo, determine the inside temperature when the outside temperature is −40.0 °C. [5]

Answer

Rate of heat conducted through the wall: $P = 0.40 \times 890 = 356$ W 1

Area of the hemispherical wall: $A = 2\pi r^2 = 2\pi(2.80)^2 = 49.3$ m^2 1

$P = kA\dfrac{\theta_{in} - \theta_{out}}{x}$ 1

$356 = (0.15)(49.3)\dfrac{\theta_{in} - (-40.0)}{0.50} \Rightarrow \theta_{in} - (-40.0) = 24.1$ °C 1

$\theta_{in} = -15.9$ °C 1

[Using the mean radius 2.55 m for the area gives about −11 °C; ECF from (i)]

5

Explanation: In the steady state the heat conducted through the snow wall each second equals 40% of the power radiated by the body. Treat the wall as a slab of thickness 0.50 m with the area of a hemisphere, $2\pi r^2$ (half of $4\pi r^2$). A temperature difference in °C is the same as in K.

(c)

(i) Describe heat transfer by convection. [3]

Answer

Heat is transferred in a fluid (liquid or gas) by the bulk movement of the fluid itself	1
Fluid near the heat source is heated, expands, becomes less dense and rises	1
Cooler, denser fluid sinks to take its place, forming a convection current	1
	3

Explanation: Convection needs a fluid that can flow, so it cannot occur in solids. The circulation is driven by the difference in density between warm and cool parts of the fluid.

(ii) Distinguish between natural and forced convections. [2]

Answer

Natural convection: the fluid moves only because of density differences caused by temperature differences (buoyancy)	1
Forced convection: the fluid is moved by an external device such as a fan or a pump	1
	2
Total for question	**15**

Explanation: The difference is what makes the fluid move. Forced convection usually transfers heat faster because the fluid moves faster, e.g. a fan cooling a computer or a pump in a car radiator.